

ORIGINAL ARTICLE

Prognostic impact of calf muscle near-infrared spectroscopy in patients with a first episode of deep vein thrombosis

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Summary. *Background:* The risk factors that affect the development of post-thrombotic syndrome (PTS) are not fully recognized, and it is difficult to reliably predict which patients are likely to develop PTS in acute phase of deep vein thrombosis (DVT). *Aims:* To investigate changes in calf muscle deoxygenated hemoglobin (HHb) levels after DVT, and to determine the indicative parameters reflecting the progression of PTS. *Methods:* Seventy-six consecutive patients with a first episode of unilateral DVT were prospectively enrolled. Clinical manifestations were categorized according to the CEAP (Clinical, Etiologic, Anatomical, and Pathophysiologic) classification, and the patients were divided into no-PTS ($C_{0-3}E_{ss}A_{s,d,p}P_{r,o}$) and PTS ($C_{4-6}E_{ss}A_{s,d,p}P_{r,o}$) groups. Near-infrared spectroscopy (NIRS) was used to measure calf muscle HHb levels at 6 months after diagnosis of DVT. The calf venous blood filling index (HHbFI) was calculated on standing, and the venous ejection index and the venous retention index (HHbRI) were then obtained after exercise. All patients were followed up for more than 24 months after the diagnosis of DVT. *Results:* Of 76 patients evaluated, 20 (26.3%) had PTS. The NIRS-derived HHbFI and HHbRI were significantly increased in patients who developed PTS in comparison with those who did not ($P = 0.04$ and $P = 0.0001$, respectively). HHbRI was significantly increased in patients with iliofemoral DVT in comparison with patients with calf DVT ($P = 0.041$). An optimal cut-off point of 2.9 for HHbRI showed the strongest ability to predict the development of PTS, with a sensitivity of 100% and a specificity of 82.1%. *Conclusions:* HHbRI as measured by NIRS is significantly increased in patients with iliofemoral DVT as compared with those with calf DVT. Furthermore, HHbRI > 2.9 is a strong predictor of the development of PTS at 6 months.

Keywords: deep vein thrombosis, near-infrared spectroscopy, post-thrombotic syndrome.

Introduction

The initial treatment of deep vein thrombosis (DVT) includes the prevention of thrombus propagation and development of pulmonary embolism, and minimizing the likelihood of recurrence. On the other hand, the need for prevention of late post-thrombotic syndrome (PTS) has emerged because of increasing awareness of its impact on patient quality of life. However, the risk factors that affect the development of PTS are not fully recognized. It is difficult to reliably predict which patients are likely to develop PTS in the acute phase of DVT.

Previous studies have demonstrated that PTS can occur many years after a venous thromboembolism [1,2]. In a phlebography study, Eichlisberger *et al.* demonstrated that 39% of patients showed chronic venous insufficiency (CVI) and that 10% developed chronic venous leg ulcers during a follow-up period of 13 years [3]. Severe PTS might be caused by gravitational venous reflux, venous obstruction, calf muscle pump malfunction, or a combination of these [1–6]. Individual anatomic venous segments in the lower extremity respond in different ways to the presence of thrombus [7]. In the late phase, the presence of reflux in the femoral and popliteal veins is considered to contribute to the development of PTS, but the role of obstruction, reflux or a combination of the two in the rest of the venous segment remains unclear [8].

Near-infrared spectroscopy (NIRS) is a non-invasive optical method for determination of tissue oxygenation and hemodynamics. Using a modification of the Lambert–Beer law [9], today it is possible to obtain quantitative values for the levels of total hemoglobin (tHb), oxygenated hemoglobin (O_2Hb), and deoxygenated hemoglobin (HHb). NIRS measurements primarily reflect changes in small arterioles, capillaries, and venules [10]. Thus, any change in HHb is presumably a consequence of a change in O_2 extraction and microvascular deoxygenation at the site of O_2 exchange, and reflects the balance between O_2 delivery and O_2 utilization in the localized region of muscle being evaluated by NIRS [11]. Recent studies have demonstrated the validity of NIRS for non-invasive measurement of tissue oxygenation in patients with peripheral

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arterial disease [12–15]. These reports graded the severity of peripheral arterial disease by assessing resaturation of the calf muscle after exercise. On the other hand, several investigators have attempted to define the severity of CVI by measuring changes in HHb, and concluded that NIRS might be useful for assessment of ambulatory venous function in patients with primary valvular insufficiency as well as PTS [16–18].

In the present study, we evaluated the distribution of DVT using duplex ultrasound, and measured the patterns of calf muscle HHb levels using NIRS. Our primary purpose was to investigate changes in calf muscle HHb levels after acute DVT, and to determine the indicative parameters reflecting the progression of PTS.

Materials and methods

Patients

Between January 2003 and February 2005, 76 consecutive patients with a first episode of unilateral DVT were prospectively evaluated using duplex ultrasound and NIRS at the Department of Plastic and Reconstructive Surgery, Tokyo Women's Medical University Hospital. There were 32 males and 44 females, ranging in age from 27 to 86 years (mean, 63 years). Basically, patients who had completed more than 24 months of follow-up after the initial episode of DVT were selected for the present study. After initial diagnosis, all patients were treated with intravenous unfractionated heparin for 5–14 days during the acute phase, adjusted to maintain the activated partial thromboplastin time at 1.5–2.5 times control [19], followed by oral warfarin for at least 8 weeks with an International Normalized Ratio level of 2–3. All patients were encouraged to ambulate, and graduated thigh-high compression stockings were applied immediately after diagnosis [20].

The patients' risk factors for DVT, including active cancer, congestive heart disease, hormone replacement therapy, immobilization, inflammatory bowel disease, surgery or trauma, previous DVT, and renal failure, were all evaluated. The thrombophilia screening in this study included protein C, protein S, antithrombin III, antiphospholipid antibody and homocysteine levels. Patients with arterial insufficiency, identified on the basis of an ankle–brachial pressure index of < 0.9 , those with recurrent DVT, fixed joints, muscle atrophy or weakness and those with limb swelling due to systemic diseases or lymphedema were excluded from the study. Factor V Leiden and prothrombin G20210A were excluded from the study, because the 76 study patients were all Japanese.

Clinical assessment

The clinical manifestations of the study patients were categorized according to the CEAP (Clinical, Etiologic, Anatomical, and Pathophysiologic) classification of reporting standards in venous disease at their last visit to the outpatient clinic [21,22]. All of the patients' lesions were classifiable into no signs of venous disease (CEAP C₀), telangiectases or reticular veins

(C₁), varicose veins (C₂), edema (C₃), changes in the skin and subcutaneous tissue (C_{4a,b}), or chronic ulcers (C₅ and C₆). Etiologic classification included secondary venous dysfunction (E_s). The anatomic sites of venous reflux included superficial (A_s), deep (A_d) and perforating (A_p) veins. As clinical signs or symptoms may result from reflux (P_r), obstruction (P_o), or both (P_{r,o}), the patients were divided into two groups: no PTS (C_{0–3}, E_s, A_{s,d,p}, P_{r,o}) and PTS (C_{4–6}, E_s, A_{s,d,p}, P_{r,o}).

Venous duplex scanning

The presence of DVT was diagnosed with venous duplex ultrasound. A color duplex scanner (LOGIQ 7 PRO; GE Yokogawa Medical Systems, Tokyo, Japan) with a 5–10-MHz transducer was used. Initially, each patient was placed supine in the reverse Trendelenburg position at 15°. Venous duplex scanning began at the external iliac vein and common femoral vein, and then moved to the femoral vein at the adductor canal. The anterior and posterior tibial veins were also recorded. The patient was then placed prone with the knee flexed at 30°, and the residual popliteal, peroneal, gastrocnemius and soleal veins were evaluated [23]. The diagnosis of DVT was based on both non-compressibility of the vein on B-mode and lack of spontaneous flow on color Doppler imaging. The diagnosis of iliofemoral DVT was established if proximal veins showed complete occlusion with venous thrombus extending proximally to the external iliac vein. The diagnosis of femoropopliteal DVT was made if proximal veins had thrombi without iliac involvement. Finally, a diagnosis of calf vein thrombosis was made if thrombus was detected only in the calf veins.

NIRS

The NIRS technique is well established, non-invasive and continuous, and operates in real time, allowing *in vivo* monitoring of tissue oxygenation and hemodynamics during certain physiologic processes [24–26]. The light source of the device (OM-200; Shimadzu Co., Kyoto, Japan) consists of three laser beams with different wavelengths (780, 805 and 830 nm), which irradiate the target area in order. According to the variations in optical density at each wavelength, O₂Hb, HHb and tHb are calculated on the basis of the Lambert–Beer law [9], which relates the variation in optical density from the initial value to the change in the concentration of hemoglobin [27]. This instrument was employed with a sensor comprising a light source and two separate detectors fixed at 2.5 cm and 4 cm from it. The sensor was placed firmly on the posterior aspect of the calf, over the medial head of the gastrocnemius muscle, with adhesive tape. Changes in the hemoglobin concentration were considered as absolute values (in $\mu\text{mol L}^{-1}$) [28].

To assess ambulatory venous function, measurements were carried out using the standard exercise protocol for air plethysmography [29]. The protocol began with the patients resting supine with the leg elevated on a foam block for 5 min. Increases in HHb were observed in the recording when the patients were instructed to adopt a standing position without

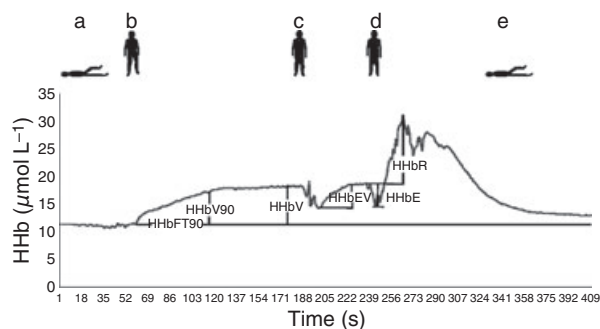


Fig. 1. Near-infrared spectroscopy (NIRS) examination. The calf venous blood filling index (HHbFI) was calculated by dividing 90% of the venous blood volume (HHbV90) by the time taken to fill 90% of the venous volume (HHbFT90). The patients were then asked to perform one tiptoe movement with weight-bearing on both legs, which produced an ejected volume (HHbEV), and then to return to the initial position, the changes in deoxygenated hemoglobin (HHb) being observed. The calf venous ejection index (HHbEI) was calculated as $\text{HHbEI} = \text{HHbEV} / \text{HHbV}$. After a new plateau had been reached, the patients were asked to perform 10 tiptoe movements, allowing venous expulsion (HHbE) and subsequent retention (HHbR). The venous retention index (HHbRI) was determined as $\text{HHbRI} = \text{HHbR} / \text{HHbE}$. a, Rest; b, standing; c, one tiptoe; d, 10 tiptoes; e, rest.

putting any weight on the leg being studied. The patients were then asked to keep still until a plateau (HHbV) had been reached. The calf venous blood filling index (HHbFI) was calculated by dividing 90% of the venous blood volume by the time taken to fill 90% of the venous volume. The patients were then asked to perform one tiptoe movement with weight-bearing on both legs, which produced an ejected volume (HHbEV), and then to return to the initial position, the changes in HHb being observed. The calf venous ejection index (HHbEI) was calculated as $\text{HHbEI} = \text{HHbEV} / \text{HHbV}$. After a new plateau had been reached, the patients were asked to perform 10 tiptoe movements to achieve venous expulsion (HHbE) and subsequent retention (HHbR). The venous retention index (HHbRI) was determined as $\text{HHbRI} = \text{HHbR} / \text{HHbE}$. The 10 tiptoe movements were completed within 10 s (Fig. 1).

Follow-up

The primary endpoint of this study was the development of PTS. In the cohort study, the follow-up protocol comprised outpatient clinic visits at 2 weeks, 1 month, 3 months, and 6 months, and every 6 months thereafter. Final clinical manifestations were evaluated more than 24 months after the diagnosis of DVT. The presence of telangiectatic leg veins, varicose veins, edema, hyperpigmentation, lipodermatosclerosis and healed or open venous ulcer was determined. In addition, the duration of oral anticoagulation was recorded.

Statistical analysis

All data were analyzed using spss software (Version 16.0; SPSS Inc., Chicago, IL, USA). Comparisons of numerical data

between groups of patients were made using Wilcoxon's non-parametric rank sum test or one-way ANOVA, followed by Fisher's PLSD *post hoc* test. Chi-squared contingency table analysis was used to evaluate differences between proportions. Receiver operating characteristic (ROC) curves were used to assess the predictive power of NIRS-derived parameters. Continuous data were expressed as mean $\pm$ standard deviation. Statistical significance was defined as $P < 0.05$.

Results

Patient characteristics

PTS was assessed at 40.4 ± 22.7 months after DVT. The baseline characteristics of the study patients are shown in Tables 1 and 2. There were 56 patients with no PTS and 20 with PTS. There was no significant difference in mean age between the groups ($P = 0.076$). Similarly, there was no significant intergroup difference in the proportion of male patients ($P = 0.082$). The mean duration of oral anticoagulation did not result in any significant difference between patients with and without PTS ($P = 0.879$). The proportion of iliofemoral DVT was significantly higher in patients who developed PTS than in patients who did not ($P = 0.043$). DVT was most common after surgery or trauma, followed by active cancer, in both groups. There were no significant differences in any of the risk factors for DVT, including thrombophilia screening.

Relationship of final clinical manifestation to NIRS-derived HHb

Changes in HHb were measured at 6 months after DVT, using the exercise protocol (Table 3). HHbFI was significantly increased in patients who developed PTS in comparison to those who did not ($P = 0.040$). Similarly, the value of HHbRI was significantly higher in patients with PTS than in those without PTS ($P = 0.0001$). There was no significant difference in the HHbEI value between patients with PTS and those without PTS ($P = 0.955$).

Relationship of initial distribution of DVT to NIRS-derived HHb

To analyze the relationship between initial distribution of DVT and NIRS-derived HHb at 6 months, the distribution of DVT was classified as iliofemoral, femoropopliteal, and calf, as described previously (Fig. 2A–C). Patients with calf DVT had increased HHbFI, but there were no significant differences in the values of HHbFI between the three categories. Among the different distributions of DVT, HHbEI was significantly reduced in patients with iliofemoral and femoropopliteal DVT as compared to those with calf DVT ($P = 0.006$ and $P = 0.001$, respectively). Similarly, HHbRI was significantly greater in patients who had iliofemoral DVT than in those with calf DVT ($P = 0.041$).

Table 1 Baseline characteristics of the study patients

	No PTS <i>n</i> = 56 patients	PTS <i>n</i> = 20 patients	<i>P</i> -value
Characteristics			
Mean age (years)	61.5 ± 15.8	65.6 ± 10.6	0.076
Male gender [no. (%)]	24 (42.9)	8 (40.0)	0.082
Duration of oral anticoagulation (months)	19.8 ± 9.8	22.0 ± 8.9	0.879
Distribution of DVT [no. (%)]			
Iliofemoral DVT	12 (21.4)	9 (45.0)	0.043
Femoropopliteal DVT	32 (57.2)	10 (50.0)	0.581
Calf DVT	12 (21.4)	1 (5.0)	0.094
Risk factors [no. (%)]			
Active cancer	10 (17.9)	2 (10.0)	0.408
Congestive heart failure	2 (3.6)	1 (5.0)	0.778
Hormone replacement therapy	7 (12.5)	2 (10.0)	0.776
Immobilization	4 (7.1)	2 (10.0)	0.684
Inflammatory bowel disease	2 (3.6)	0 (0)	0.392
Surgery or trauma	21 (37.5)	7 (35.0)	0.180
Previous DVT	3 (5.4)	2 (10.0)	0.472
Renal failure	2 (3.6)	0 (0)	0.392
Stroke	6 (10.7)	4 (20.0)	0.292
Thrombophilia	6 (10.7)	2 (10.0)	0.929
Protein C deficiency	3 (5.4)	1 (5.0)	0.951
Protein S deficiency	2 (3.6)	1 (5.0)	0.778
Antithrombin III deficiency	1 (1.8)	0 (0)	0.548
Positive antiphospholipid syndrome	1 (1.8)	0 (0)	0.548
Hyperhomocysteinemia	0 (0)	0 (0)	NS

DVT, deep vein thrombosis; NS, not significant; PTS, post-thrombotic syndrome.

Table 2 The number of patients in each CEAP (Clinical, Etiologic, Anatomic, and Pathophysiologic) clinical classification

Class	Signs and symptoms	No. of patients
C ₀	No visible or palpable signs of venous disease	43
C ₁	Telangiectasis or reticular veins	4
C ₂	Varicose veins; distinguished from reticular veins by a diameter of 3 mm or more	4
C ₃	Edema	5
C _{4a}	Pigmentation or eczema	13
C _{4b}	Lipodermatosclerosis or atrophic blanche	5
C ₅	Healed venous ulcer	2
C ₆	Active venous ulcer	0

Table 3 Relationship of final clinical manifestation to near-infrared spectroscopy-derived deoxygenated hemoglobin

	No PTS <i>n</i> = 56 patients	PTS <i>n</i> = 20 patients	<i>P</i> -value
On standing			
HHbFI (μmol L ⁻¹)	0.05 ± 0.03	0.15 ± 0.14	0.040
During exercise			
HHbEI	0.37 ± 0.29	0.32 ± 0.18	0.955
HHbRI	2.68 ± 3.86	11.14 ± 9.16	0.0001

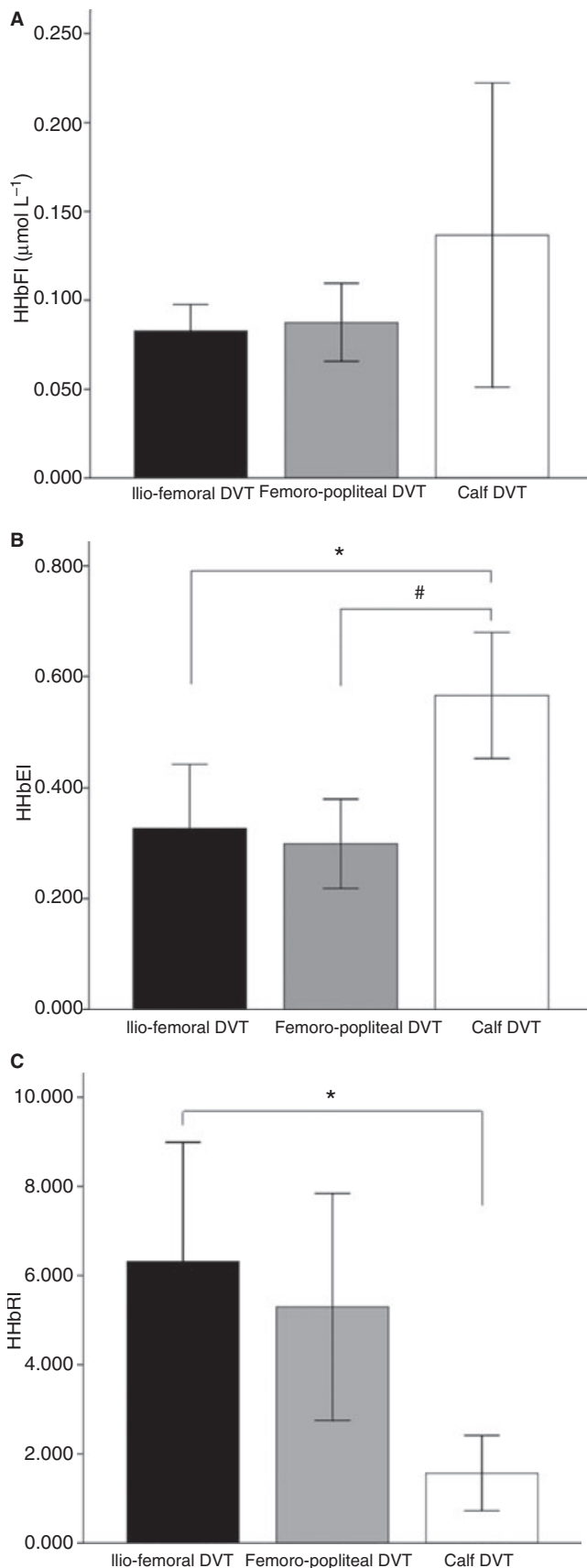
HHbEI, venous ejection index; HHbFI venous blood filling index; HHbRI, venous retention index; PTS, post-thrombotic syndrome.

Ability of NIRS-derived HHb to predict PTS

With NIRS-derived parameters as a unit of analysis, separate ROC curve analysis was conducted, and the most suitable cut-off points, with the highest accuracy, minimal false negativity and minimal false positivity for discrimination between the two groups, were then identified (Fig. 3A–C). First, an optimal cut-off point of 2.9 for HHbRI showed the strongest ability to predict the development of PTS, with a sensitivity of 100%, a specificity of 82.1%, a positive predictive value of 66.7%, and a negative predictive value of 100% [area under the ROC curve = 0.91, 95% confidence interval (CI) 0.82–0.96, *P* = 0.0001]. Second, the clinical discrimination performance was significantly reduced when a single HHbFI cut-off point of 0.13 was used (area under the ROC curve = 0.65, 95% CI 0.53–0.75, *P* = 0.048). Finally, an HHbEI cut-off of 0.32 did not have any power to discriminate between patients with and without PTS (area under the ROC curve = 0.49, 95% CI 0.38–0.61, *P* = 0.925).

Discussion

This study investigated the relationship between changes in calf muscle HHb and clinical outcome of DVT. Our main findings were that: (i) calf muscle HHbFI and HHbRI measured at 6 months were significantly increased in patients who developed PTS more than 24 months after DVT; (ii) NIRS-derived HHbEI was significantly decreased in patients with proximal DVT as compared with those with calf DVT; (iii)



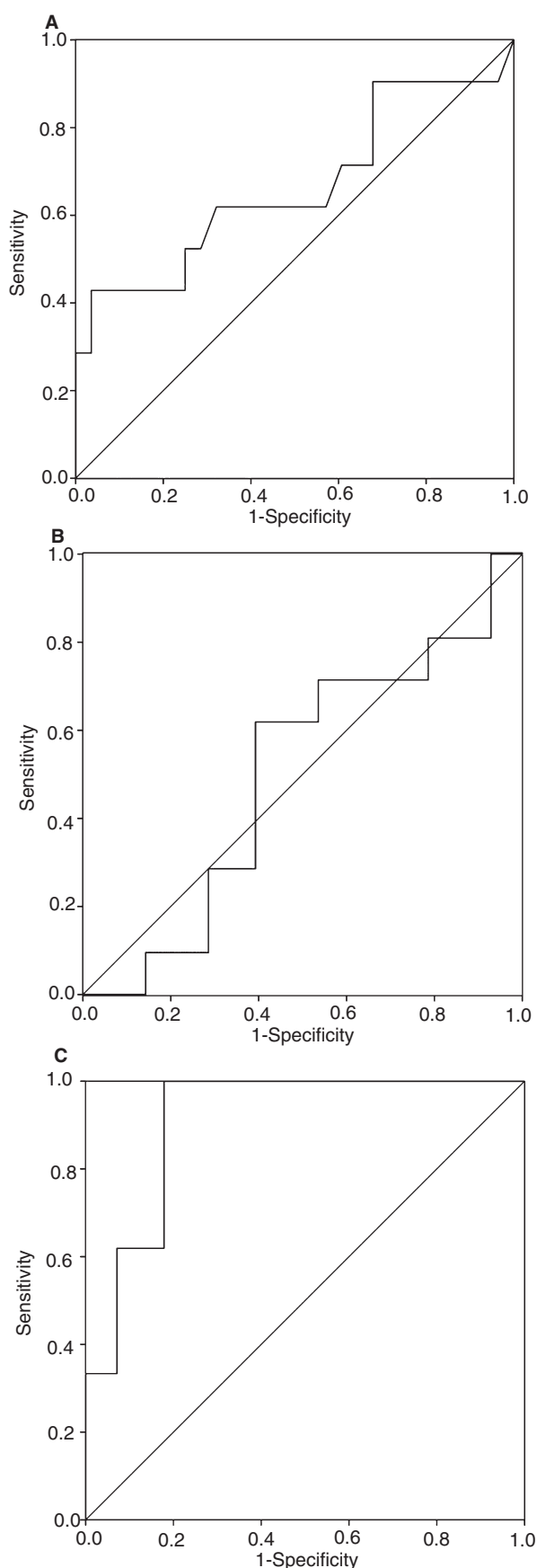
NIRS-derived HHbRI was significantly increased in patients with iliofemoral DVT as compared with those with calf DVT; and (iv) an optimal cut-off point of 2.9 for HHbRI showed the strongest ability to predict the development of PTS, with a sensitivity of 100%, a specificity of 82.1%, a positive predictive value of 66.7%, and a negative predictive value of 100%.

The incidence of PTS varies from 20% to 50%, with 5–10% of the patients developing severe PTS after the episode of DVT [30–33]. The symptoms of PTS appear within the first 2 years after DVT in the majority of patients [31]. A recent large prospective cohort study showed that 43% of first episodes of DVT lead to PTS within 3 years [34]. However, severe PTS with venous ulcers was seen in only < 2%. In the present cohort, two (2.6%) patients developed disabling venous ulceration during the follow-up period.

The factors that affect the risk of developing PTS are not well recognized. Increasing age has been proven to be associated with a higher risk of DVT. However, there appears to be no definite association with PTS [30,35,36]. There is no conclusive evidence that men are at higher risk of developing PTS after DVT [34,35]. The role of thrombophilia work-up with regard to the risk of PTS is also uncertain. Stain *et al.* found that the presence of factor V Leiden, prothrombin gene mutation or an elevated FVIII level were not predictive of the development of PTS [34]. In contrast, Kahn *et al.* showed that the presence of FV Leiden or prothrombin mutation was an independent predictor of a lower risk of PTS [36]. Although the risk of recurrent DVT increases after discontinuation of anticoagulation, the duration of anticoagulant therapy within the therapeutic range did not appear to be predictive of the risk of developing PTS [34].

There are conflicting results with regard to whether proximal or distal DVT is more likely to be associated with advanced symptoms of PTS. Several studies have compared the extent and distribution of DVT with the symptoms of PTS. One study showed that the extent of DVT was similar in limbs with PTS and those without PTS [37], whereas another study found that PTS was more likely to develop in patients with multisegment DVT than in those with isolated-segment DVT [38]. The location of the thrombus may also affect the progression of PTS. Some studies have reported no relationship between the distribution of DVT and the development of PTS [32,39]. In

Fig. 2. Relationship of initial distribution of deep vein thrombosis (DVT) to near-infrared spectroscopy-derived deoxygenated hemoglobin. (A) There were no significant differences in the values of calf venous blood filling index (HHbFI) between the three categories. (B) Among the different anatomic distributions of DVT, the venous ejection index (HHbEI) was significantly reduced in patients with iliofemoral and femoropopliteal DVT as compared with those with calf DVT. * $P = 0.006$ compared with calf DVT; $P = 0.001$ as compared with calf DVT. (C) The venous retention index (HHbRI) was significantly greater in patients who had iliofemoral DVT than in those with calf DVT. * $P = 0.041$ as compared with calf DVT.



other studies, the risk of PTS was reportedly higher in patients with proximal rather than distal DVT [6,34]. Proximal DVT confers more than a two-fold increased likelihood of PTS [34]. Furthermore, patients with acute iliofemoral DVT will suffer the most severe post-thrombotic sequelae [40]. Cases of calf DVT may show faster recanalization, due to the function of the calf muscle pump, and the risk of PTS increases only in patients who have calf DVT with proximal DVT. These findings are in accord with those of our present study.

Previous venous thrombosis may also affect venous hemodynamics. Severe post-thrombotic symptoms could be caused by either obstruction or reflux in the axial deep venous system, or a combination of the two. Johnson *et al.* found that only 3% of limbs with PTS were normal, 18% had reflux alone, 15% had obstruction alone, and 65% had both reflux and obstruction, and concluded that a combination of residual reflux and residual partial or complete obstruction in the major deep veins was more likely to be associated with the development of PTS [37]. On the other hand, Killewich *et al.* showed that reflux in the deep veins was the main contributor to the severity of PTS [41]. Haenen *et al.* also found that reflux in the deep proximal veins contributed significantly to the development of PTS [42]. We have studied venous obstruction and measured quantitative reflux by duplex scanning in a population of patients who completed 6 years of follow-up after DVT, and reported that the presence of reflux as reflected by elevated peak reflux velocity in the popliteal ($> 25.4 \text{ cm s}^{-1}$) and femoral ($> 24.5 \text{ cm s}^{-1}$) veins was an independent predictor of advanced PTS [43]. These reports show that axial reflux with increased peak reflux velocity in proximal deep veins plays an important role in the development of severe PTS symptoms.

The first observations of tissue oxygenation using NIRS were made by Jöbsis [44]. By choosing an accurate distance between source and detector, NIRS can provide direct measurements for any muscle of interest, and over the last two decades, many groups have applied this technique for the evaluation of muscle metabolism [13–15,45–47]. Interestingly, the present study demonstrated that significant increases in both HHbFI and HHbRI at 6 months were correlated with the development of PTS. Furthermore, HHbRI showed the strongest ability to predict the development of PTS. The

Fig. 3. Ability of near-infrared spectroscopy-derived deoxygenated hemoglobin to predict post-thrombotic syndrome (PTS). (A) Clinical discrimination performance was significantly reduced when a single calf venous blood filling index cut-off point of 0.13 was used [area under the receiver operating characteristic (ROC) curve = 0.65, 95% confidence interval (CI) 0.53–0.75, $P = 0.048$]. (B) A venous ejection index cut-off of 0.32 did not have any power to discriminate between patients with and without PTS (area under the ROC curve = 0.49, 95% CI 0.38–0.61, $P = 0.925$). (C) An optimal cut-off point of 2.9 for venous retention index showed the strongest ability to predict the development of PTS, with a sensitivity of 100%, a specificity of 82.1%, a positive predictive value of 66.7%, and a negative predictive value of 100% (area under the ROC curve = 0.91, 95% CI 0.82–0.96, $P = 0.0001$).

significant increase in HHb retention suggests venous obstruction and increased venous reflux after DVT, with an increased risk of PTS. In addition, HHb ejection was significantly decreased in patients with proximal DVT as compared with those with calf DVT, and HHb retention was significantly higher in patients with iliofemoral DVT than in those with calf DVT. These factors might reflect diminished calf muscle pump function as a result of pre-existing venous obstruction, venous reflux, or a combination of the two, leading to higher ambulatory venous pressure and development of PTS.

Despite the considerable incidence of PTS encountered in clinical practice, objective assessment of its severity is still difficult. First, venous abnormalities determined by duplex scanning at different levels do not reflect overall venous hemodynamics, and duplex imaging, which can offer more physiologic information, is difficult to obtain during exercise. Second, data obtained by air plethysmography (APG) could be used to evaluate ambulatory venous function, but APG parameters do not correctly reflect data obtained in the presence of proximal occlusion. In contrast, venous occlusion does not affect NIRS-derived parameters. In the past, T. Yamaki tried to correlate APG-derived parameters with NIRS-derived parameters, but found there were no significant relationships between venous filling index and HHbFI, ejection fraction and HHbEI, or residual volume fraction and HHbRI (unpublished data). The estimated cost of NIRS is \$69 (6700 JPY) per procedure, and that of APG is approximately \$13 (1300 JPY) per procedure. However, this cost estimation does not take into account additional related direct and indirect costs.

Our study had some potential limitations. First, we excluded patients with recurrent DVT, even though it is recognized as a high-risk factor for PTS. Because NIRS-derived parameters will change after the development of recurrent DVT, the relationship between the initial distribution of DVT and these parameters might have been concealed. Second, we did not analyze potential risk factors using multivariate analysis, mainly because of the limited sample size, and this may represent a potential type II statistical error. Nevertheless, our study clearly demonstrated the significance of NIRS-derived HHbRI for prediction of PTS at 6 months in patients with a first episode of DVT.

In conclusion, the present results suggest that HHbFI and HHbRI as measured by NIRS may be promising parameters for prediction of PTS after DVT. Calf muscle HHbEI as measured by NIRS is significantly reduced in patients who have had proximal DVT in comparison with those who have had calf DVT. HHbRI is significantly increased in patients with iliofemoral DVT in comparison with those with calf DVT. Furthermore, an optimal cut-off point of 2.9 for HHbRI provides a strong capability to predict advanced symptoms of PTS. These results appear to reflect the possibility that increased calf muscle deoxygenation during exercise suggests future venous malfunction in patients with a first episode of DVT.

Disclosure of Conflict of Interests

The authors state that they have no conflict of interest.

References

- 1 Franzeck UK, Schalch I, Jager KA, Schneider E, Grimm J, Bollinger A. Prospective 12-year follow-up study of clinical and hemodynamic sequelae after deep vein thrombosis in low-risk patients (Zurich study). *Circulation* 1996; **93**: 74–9.
- 2 Johnson BF, Manzo RA, Bergelin RO, Strandness DE Jr. The site of residual abnormalities in the leg veins in long-term follow-up after deep vein thrombosis and their relationship to the development of the post-thrombotic syndrome. *Int Angiol* 1996; **15**: 14–19.
- 3 Eichlisberger R, Frauchiger B, Widmer M, Widmer L, Jager K. Late sequelae of deep vein thrombosis: a 13-year follow-up of 223 patients. *VASA* 1994; **23**: 234–43.
- 4 Janssen MCH, Wollersheim H, van Asten WNJC, de Rooij MJM, Novakova IRO, Thien T. The post-thrombotic syndrome: a review. *Phlebology* 1996; **11**: 86–94.
- 5 Haenen JH, Janssen MCH, van Langen H, van Asten WNJC, Wollersheim H, Heystraten FMJ, Skotnicki SH, Thien T. Duplex ultrasound in the hemodynamic evaluation of the late sequelae of deep venous thrombosis. *J Vasc Surg* 1998; **27**: 472–8.
- 6 Meissner MH, Caps MT, Zierler BK, Polissar N, Bergelin RO, Manzo RA, Strandness DE Jr. Determinants of chronic venous disease after acute deep venous thrombosis. *J Vasc Surg* 1998; **28**: 826–33.
- 7 Yamaki T, Nozaki M. Patterns of venous insufficiency after an acute deep vein thrombosis. *J Am Coll Surg* 2005; **201**: 231–8.
- 8 Shull KC, Nicolaides AN, Fernandes e Fernandes J, Miles C, Horner J, Needham T, Cooke ED, Eastcott FH. Significance of popliteal reflux in relation to ambulatory venous pressure and ulceration. *Arch Surg* 1979; **114**: 1304–6.
- 9 Beer A. Versus der Absorptions-Verhältnisse des Cordietes für rothes Licht zu bestimmen. *Ann Physik Chem* 1851; **84**: 37–52.
- 10 Boushel R, Langberg H, Olesen J, Gonzales-Alonzo J, Bulow J, Kjaer M. Monitoring tissue oxygen availability with near infrared spectroscopy (NIRS) in health and disease. *Scand J Med Sci Sports* 2001; **11**: 213–22.
- 11 DeLorey DS, Kowalchuk JM, Paterson DH. Relationship between pulmonary O₂ uptake kinetics and muscle deoxygenation during moderate-intensity exercise. *J Appl Physiol* 2003; **95**: 113–20.
- 12 Cheattle TR, Potter LA, Cope M, Delpy DT, Coleridge-Smith PD, Scurr JH. Near-infrared spectroscopy in peripheral vascular disease. *Br J Surg* 1991; **78**: 405–8.
- 13 Komiyama T, Shigematsu H, Yasuhara H, Muto T. An objective assessment of intermittent claudication by near-infrared spectroscopy. *Eur J Vasc Surg* 1994; **8**: 294–6.
- 14 Komiyama T, Shigematsu H, Yasuhara H, Muto T. Near-infrared spectroscopy grades the severity of intermittent claudication in diabetics more accurately than ankle pressure measurement. *Br J Surg* 2000; **87**: 459–66.
- 15 Egun A, Fapooq V, Torella F, Cowley R, Thorniley MS, McCollum CN. The severity of muscle ischemia during intermittent claudication. *J Vasc Surg* 2002; **36**: 89–93.
- 16 Hosoi Y, Yasuhara H, Shigematsu H, Aramoto H, Komiyama T, Muto T. A new method for the assessment of venous insufficiency in primary varicose veins using near-infrared spectroscopy. *J Vasc Surg* 1997; **26**: 53–60.
- 17 Hosoi Y, Yasuhara H, Shigematsu H, Komiyama T, Onozuka A, Muto T. Influence of popliteal venous thrombosis on subsequent ambulatory venous function measured by near-infrared spectroscopy. *Am J Surg* 1999; **177**: 111–16.

- 18 Yamaki T, Nozaki M, Sakurai H, Takeuchi M, Soejima K, Kono T. The utility of quantitative calf muscle near-infrared spectroscopy in the follow-up of acute deep vein thrombosis. *J Thromb Haemost* 2006; **4**: 800–6.
- 19 Hirsh J, Anand SS, Halperin JL, Fuster V. Guide to anticoagulant therapy: heparin: a statement for healthcare professionals from the American Heart Association. *Circulation* 2001; **103**: 2994–3018.
- 20 Partsch H, Blättler W. Compression and walking versus bed rest in the treatment of proximal deep venous thrombosis with low molecular weight heparin. *J Vasc Surg* 2000; **32**: 861–9.
- 21 Porter JM, Moneta GL. International Consensus Committee on Chronic Venous Disease. Reporting Standard in venous disease: an update. *J Vasc Surg* 1995; **2**: 635–45.
- 22 Eklöf B, Rutherford RB, Bergan JJ, Carpentier PH, Gloviczki P, Kistner RL, Meissner MH, Moneta GL, Myers K, Padberg FT, Perrin M, Ruckley V, Coleridge-Smith P, Wakefield TW. Revision of the CEAP classification for venous disorders: consensus statement. *J Vasc Surg* 2004; **40**: 1248–52.
- 23 Miller N, Satin R, Tousignant L, Sheiner MN. A prospective study comparing duplex scan and venography for the diagnosis of lower-extremity deep vein thrombosis. *Cardiovasc Surg* 1996; **4**: 505–8.
- 24 Delpy DT, Cope M, van der Zee P, Arrige S, Wray S, Wyatt J. Estimation of optical path length through tissue from direct time of flight measurement. *Phys Med Biol* 1988; **33**: 1433–42.
- 25 Binzoni T, Quaresima V, Ferrari M, Hiltbrand E, Cerretelli P. Human calf microvascular compliance measured by near-infrared spectroscopy. *J Appl Physiol* 2000; **88**: 369–72.
- 26 Grassi B, Quaresima V, Marconi C, Ferrari M, Cerretelli P. Blood lactate accumulation and muscle deoxygenation during incremental exercise. *J Appl Physiol* 1999; **87**: 348–55.
- 27 Kawaguchi T, Urayama O, Konishi M, Nishiyama T, Iida T. Orthostatic hypotension in elderly persons during passive standing: a comparison with young persons. *J Gerontol* 2001; **56A**: M273–8.
- 28 Komiyama T, Quaresima V, Shigematsu H, Ferrari M. Comparison of two spatially resolved near-infrared photometers in the detection of tissue oxygen saturation: poor reliability at very low oxygen saturation. *Clin Sci* 2001; **101**: 715–18.
- 29 Nicolaides AN, Christopoulos D. Quantification of venous reflux and outflow obstruction with air plethysmography. In: Bernstein EF, ed. *Vascular Diagnosis*. St Louis, MO: Mosby, 1993: 915–21.
- 30 Schulman S, Lindmarker P, Holmström M, Lärfaars G, Carlsson A, Nicol P, Svensson E, Ljungberg B, Vierung S, Nordlander S, Leijed B, Jahed K, Hjørth M, Linder O, Beckman M. Post-thrombotic syndrome, recurrence, and death 10 years after the first episode of venous thromboembolism treated with warfarin for 6 weeks or 6 months. *J Thromb Haemost* 2006; **4**: 734–42.
- 31 Kahn SR, Ginsberg JS. Relationship between deep venous thrombosis and the post-thrombotic syndrome. *Arch Intern Med* 2004; **164**: 17–26.
- 32 Prandoni P, Lensing AW, Cogo A, Cuppini S, Villalta S, Carta M, Cattelan AM, Polistena P, Bernardi E, Prins MN. The long-term clinical course of acute deep vein thrombosis. *Ann Intern Med* 1996; **125**: 1–7.
- 33 Prandoni P, Lensing AWA, Prins MH, Frulla M, Marchiori A, Bernardi E, Tormene D, Mosena L, Pagnan A, Girolami A. Below-knee elastic compression stockings to prevent the post-thrombotic syndrome: a randomized, controlled trial. *Ann Intern Med* 2004; **141**: 249–56.
- 34 Stain M, Schönauer V, Minar E, Bialonczyk C, Hirschl M, Weltermann A, Kyrle PA, Eichinger S. The post-thrombotic syndrome: risk factors and impact on the course of thrombotic disease. *J Thromb Haemost* 2005; **3**: 2671–6.
- 35 van Dongen CJ, Prandoni P, Frulla M, Marchiori A, Prins MH, Hutten BA. Relation between quality of anticoagulant treatment and the development of the post-thrombotic syndrome. *J Thromb Haemost* 2005; **3**: 939–42.
- 36 Kahn SR, Kearon C, Julian JA, Mackinnon B, Kovacs MJ, Wells P, Crowther MA, Anderson DR, van Ngyen P, Demers C, Soymoss S, Kassis J, Geerts W, Rodger M, Hambleton J, Ginsberg JS for the Extended Low-intensity Anticoagulation for Thrombo-embolism (ELATE) investigators. Predictors of the post-thrombotic syndrome during long-term treatment of proximal deep vein thrombosis. *J Thromb Haemost* 2005; **3**: 718–23.
- 37 Johnson BF, Manzo RA, Bergelin RO, Strandness DE Jr. Relationship between changes in the deep venous system and the development of the post-thrombotic syndrome after an acute episode of lower limb deep vein thrombosis: a one- to six-year follow-up. *J Vasc Surg* 1995; **21**: 307–13.
- 38 Labropoulos N, Waggoner T, Sammis W, Samali S, Pappas PJ. The effect of venous thrombus location and extent on the development of post-thrombotic signs and symptoms. *J Vasc Surg* 2008; **48**: 407–12.
- 39 Prandoni P, Villalta S, Bagatella P, Rossi L, Marchiori A, Piccioli A, Bernardi E, Girolami B, Simioni P, Girolami A. The clinical course of deep-vein thrombosis. Prospective long-term follow-up of 528 symptomatic patients. *Haematologica* 1997; **82**: 423–8.
- 40 Delis KT, Bountouroglou D, Mansfield AO. Venous claudication in iliofemoral thrombosis: long-term effect on venous hemodynamics, clinical status, and quality of life. *Ann Surg* 2004; **239**: 118–26.
- 41 Killewich LA, Martin R, Cramer M, Beach KW, Strandness DE. An objective assessment of the physiologic changes in the post-thrombotic syndrome. *Arch Surg* 1985; **120**: 424–6.
- 42 Haenen JH, Janssen MCH, van Langen H, van Asten WJNC, Wollersheim H, van't Hof MA, Skotnicki SH, Thien T. The post-thrombotic syndrome in relation to venous hemodynamics, as measured by means of duplex scanning and strain-gauge plethysmography. *J Vasc Surg* 1999; **29**: 1071–6.
- 43 Yamaki T, Nozaki M, Sakurai H, Takeuchi M, Kono T, Soejima K. High peak reflux velocity in the proximal deep veins is a strong predictor of advanced post-thrombotic sequelae. *J Thromb Haemost* 2007; **5**: 305–12.
- 44 Jöbsis FF. Noninvasive infrared monitoring of cerebral and myocardial oxygen sufficiency and circulatory parameters. *Science* 1977; **198**: 1264–7.
- 45 Chance B, Leigh J, Clark B, Maris J, Kent J, Nioka S. Control of oxidative metabolism and oxygen delivery in human skeletal muscle: a steady-state analysis of work/energy cost transfer function. *Proc Natl Acad Sci USA* 1985; **82**: 8384–8.
- 46 Belardinelli R, Barstow T, Porszasz J, Wasserman K. Changes in skeletal muscle oxygenation during incremental exercise measured with near infrared spectroscopy. *Eur J Appl Physiol* 1995; **70**: 487–92.
- 47 De Blasi RA, Cope M, Elwell C, Safoue F, Ferrari M. Noninvasive measurement of human forearm oxygen consumption by near infrared spectroscopy. *Eur J Appl Physiol* 1993; **67**: 20–5.